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(57)

ABSTRACT

A dehumidifier is provided that may include a base with an external air intake on one side thereof; a case disposed on an upper surface of the base and having a suction grille formed on a rear surface thereof; a top cover disposed on an upper surface of the case and having a discharge port; a discharge cover that opens and closes the discharge port; a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends from a lower surface of the first portion and disposed on the base; a water tank disposed on the upper surface of the base and that stores condensed water generated during a dehumidification process; a compressor disposed on the upper surface of the base spaced apart from the water tank; a fan module disposed on the first portion and having an outlet that communicates with the discharge port of the top cover; a condenser mounted on the first portion at an intake side of the fan module; an evaporator mounted on the first portion behind the condenser; and a control box coupled to the second portion and disposed directly above the external air intake. The fan module may include a fan housing disposed on the first portion, and a fan accommodated in the fan housing. A dripping hole may be formed in the fan housing.

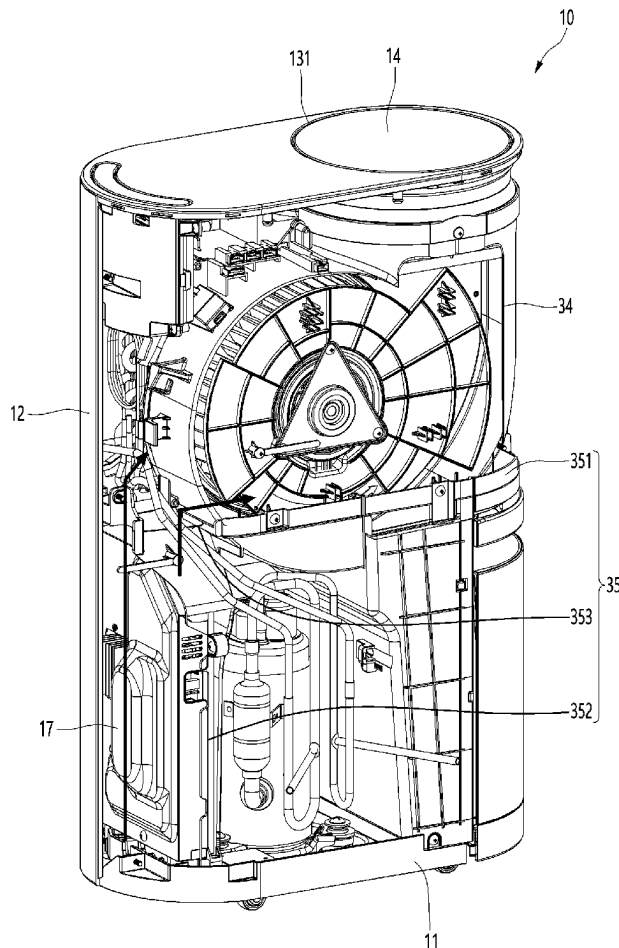


FIG. 1

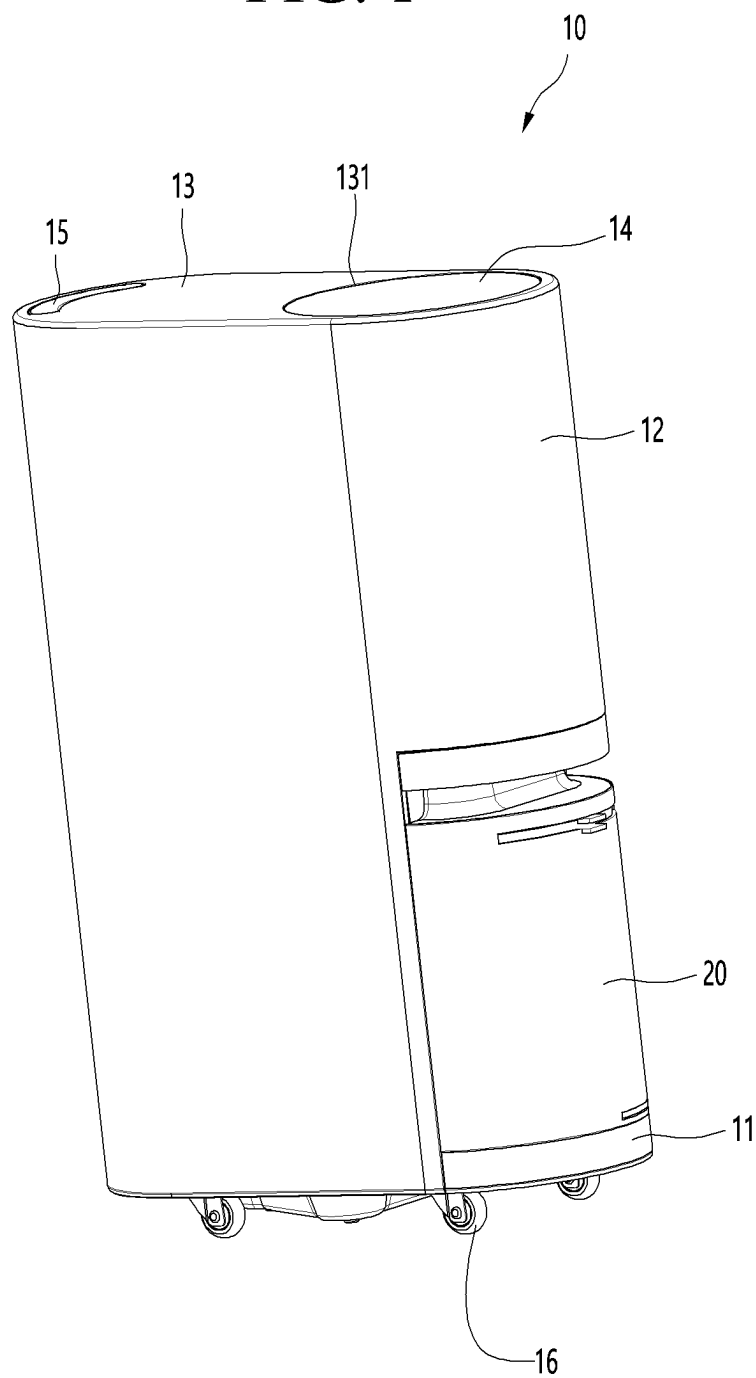


FIG. 2

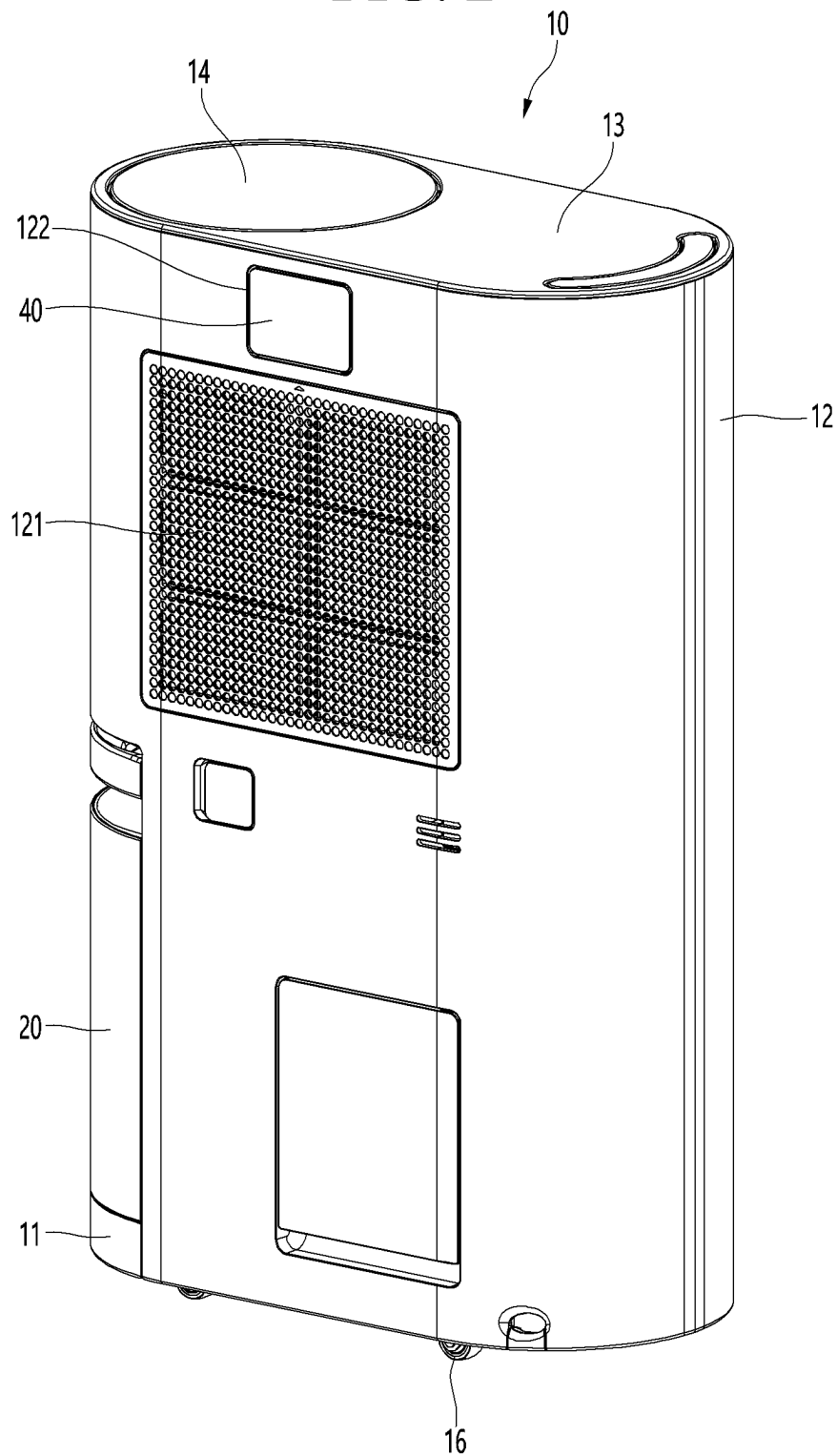


FIG. 3

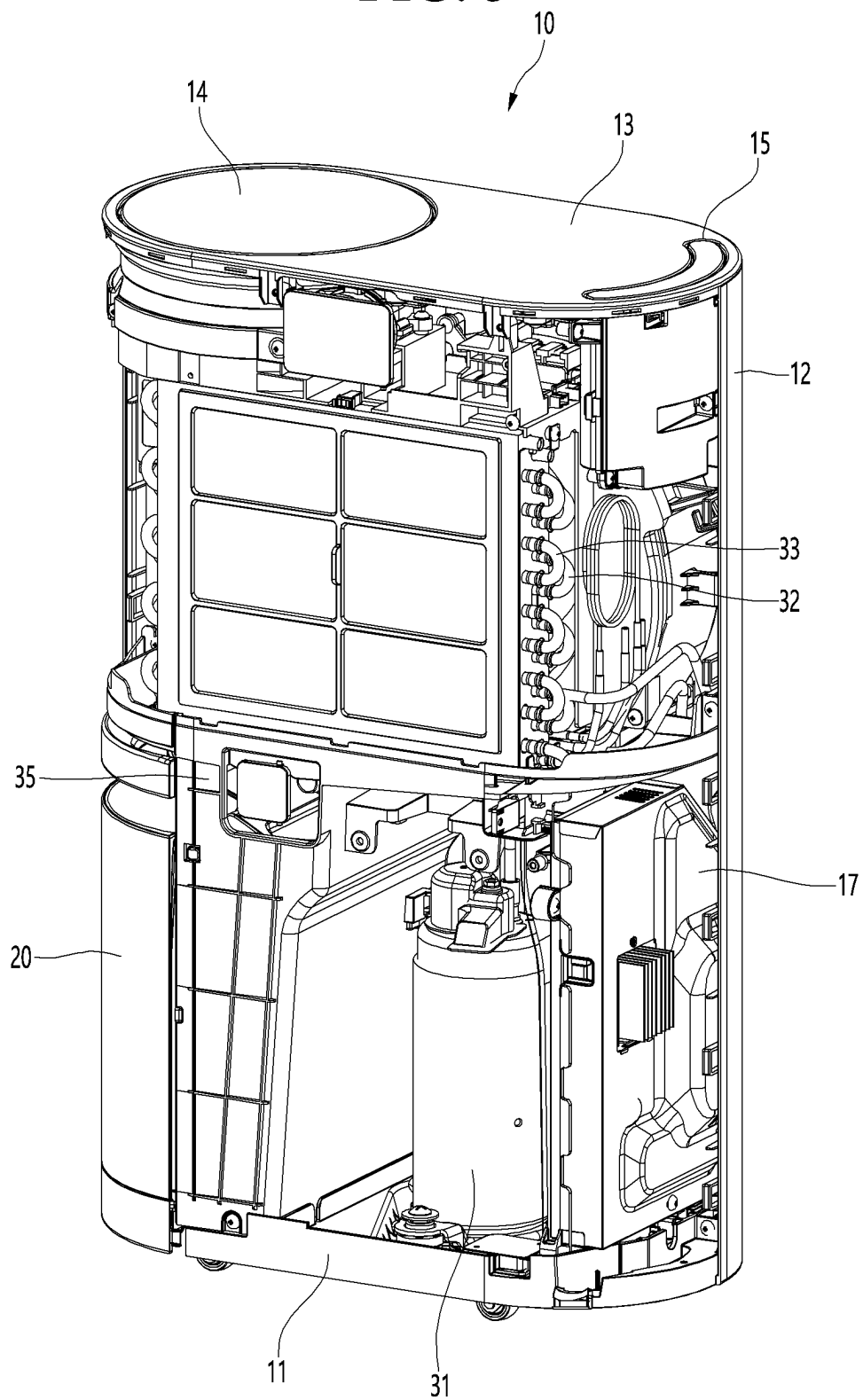


FIG. 4

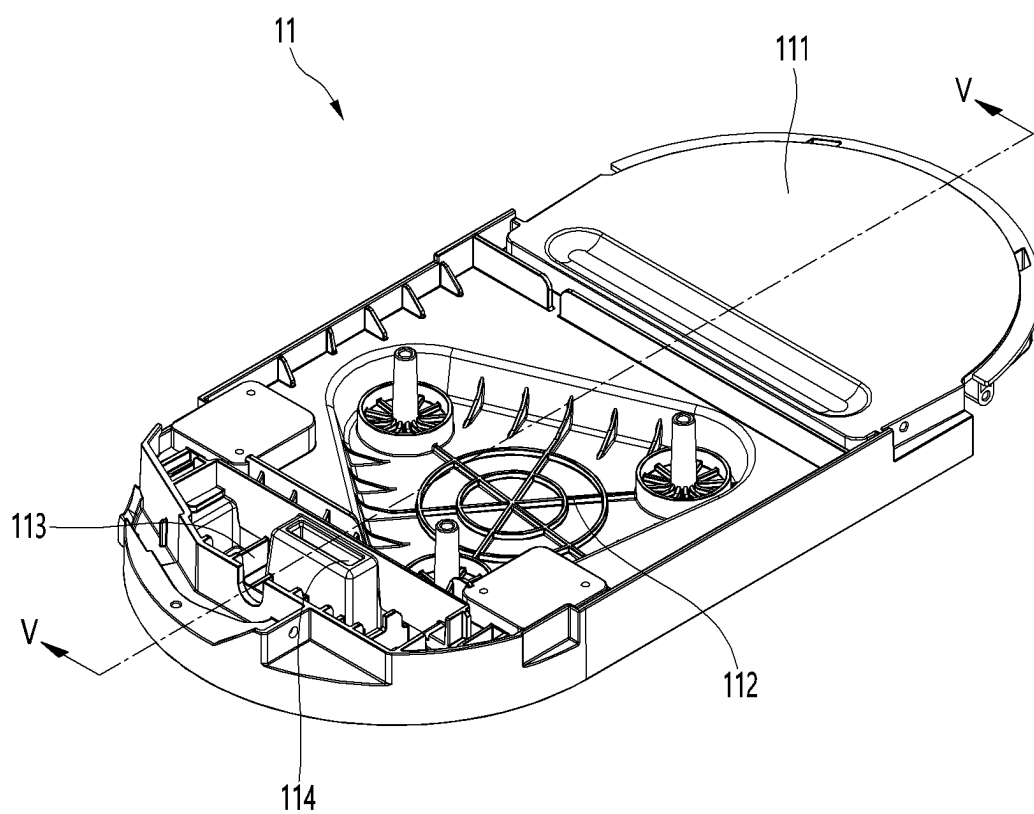


FIG. 5

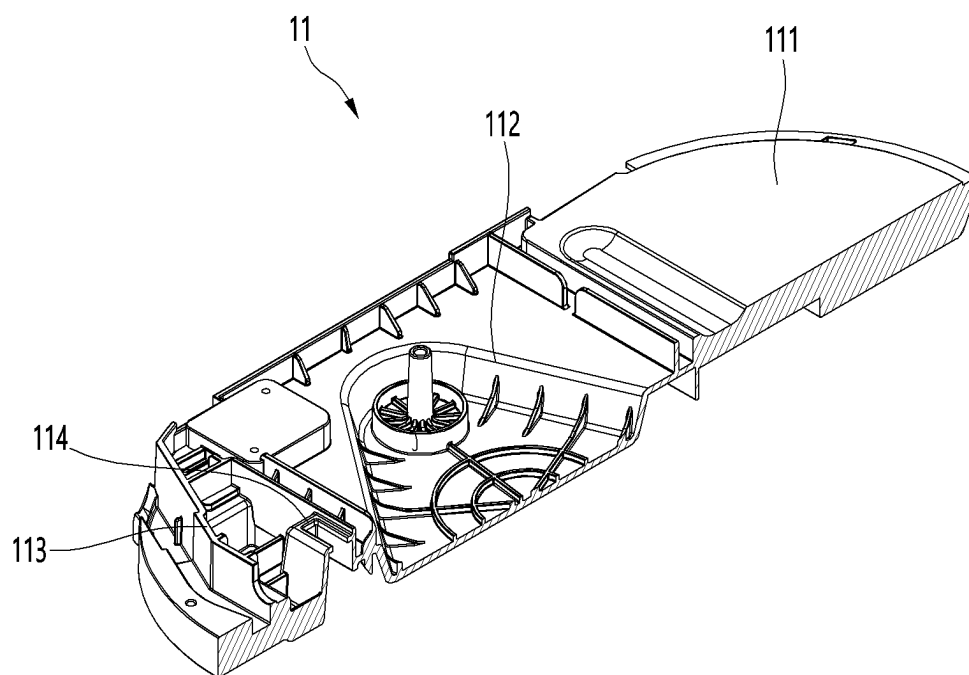


FIG. 6

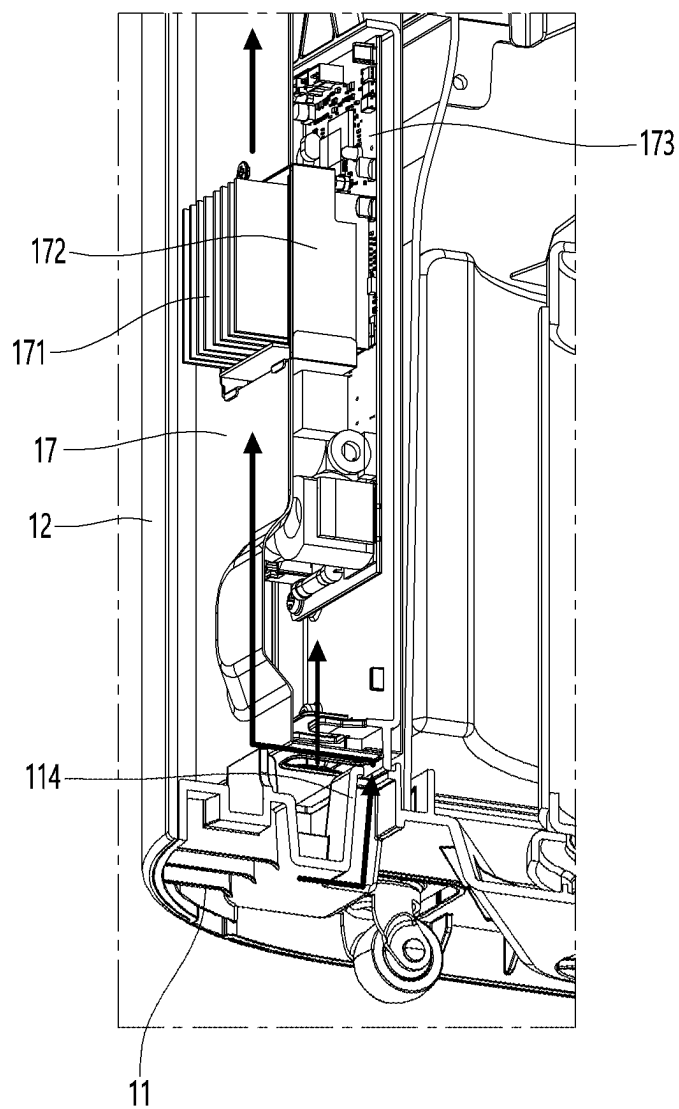


FIG. 7

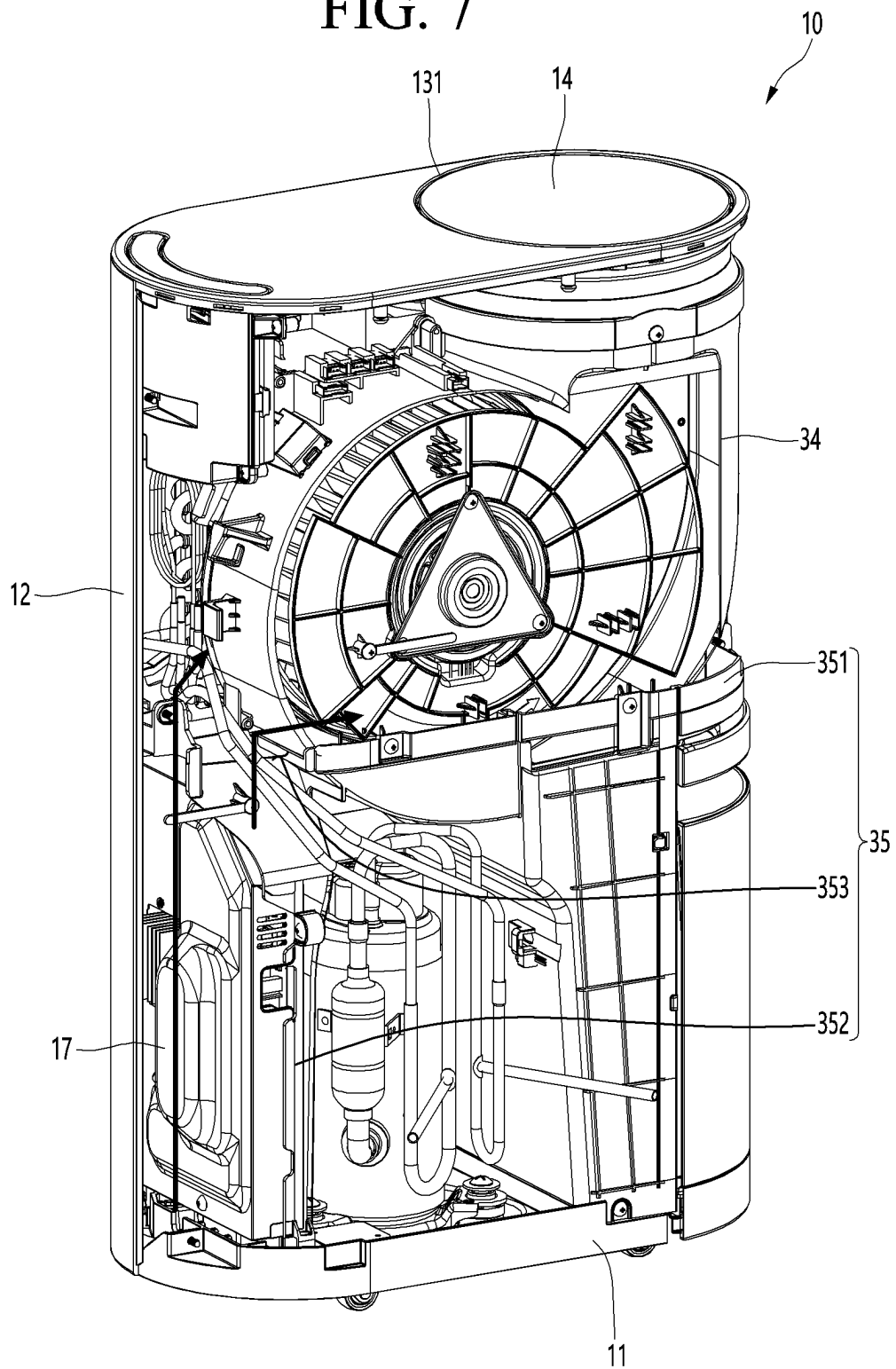
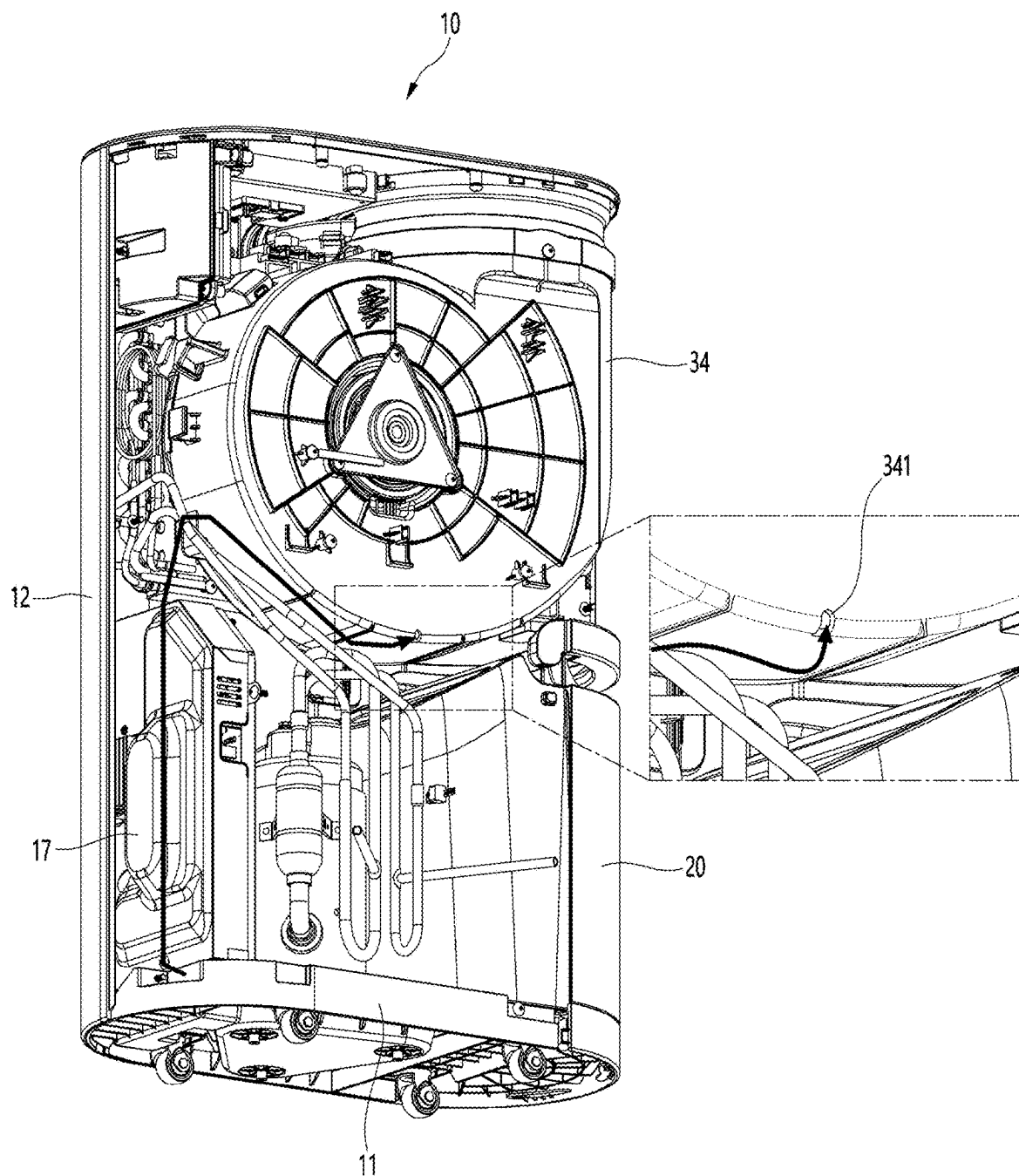


FIG. 8



DEHUMIDIFIER

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefits of priority to Korean Patent Application No. 10-2024-0033872, filed in Korea on Mar. 11, 2024, the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

[0002] A dehumidifier is disclosed herein.

2. Background

[0003] A dehumidifier is a device that lowers a humidity in a space in which the dehumidifier is installed by suctioning in air, removing moisture contained in the air, and discharging the dehumidified air back into the room. In general, dehumidifiers are widely used to lower the humidity of indoor air in humid summer or to quickly dry laundry.

[0004] Dehumidifiers are divided into a desiccant type and a condensation type depending on a dehumidification method. In the condensation type, a refrigerant cycle is installed inside of the dehumidifier, air suctioned in by a fan passes through an evaporator to condense moisture contained in the air, and then passes through the condenser and is discharged back into a room in a heated state.

[0005] In the case of an inverter dehumidifier, electrical components that generate a lot of heat are installed inside of a control box, and it is necessary to quickly cool the electrical components that generate a lot of heat. However, conventional dehumidifiers do not provide a separate cooling passage structure to cool the control box, so there is a high risk of overheating the control box.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

[0007] FIG. 1 is a front perspective view of a dehumidifier according to an embodiment;

[0008] FIG. 2 is a rear perspective view of the dehumidifier of FIG. 1;

[0009] FIG. 3 is a schematic diagram showing an internal configuration of the dehumidifier according to an embodiment;

[0010] FIG. 4 is a perspective view of a base of a housing of the dehumidifier according to an embodiment;

[0011] FIG. 5 is a longitudinal cross-sectional view of the base, taken along line V-V in FIG. 4; and

[0012] FIGS. 6 to 8 are schematic diagrams showing a flow of indoor air flowing in through an external air inlet of the base.

DETAILED DESCRIPTION

[0013] Hereinafter, a dehumidifier according to an embodiment will be described with reference to the drawings.

[0014] FIG. 1 is a front perspective view of a dehumidifier according to an embodiment. FIG. 2 is a rear perspective view of the dehumidifier of FIG. 1. FIG. 3 is a schematic

diagram showing an internal configuration of the dehumidifier according to an embodiment.

[0015] Referring to FIGS. 1 to 3, dehumidifier 10 according to an embodiment may have a slim shape with a relatively thin thickness in a frontward-to-backward direction. The dehumidifier 10 may have a thickness extending in the frontward-to-backward direction, a width extending in a leftward-rightward or lateral direction, and a height extending in a vertical direction.

[0016] The dehumidifier 10 may include a housing that forms an exterior thereof. The dehumidifier 10 may further include a dehumidification module installed inside of the housing. The dehumidifier 10 may further include a water tank 20 that stores condensate generated during the dehumidification process.

[0017] The housing may include a base 11, a case 12, and a top cover 13. The base 11 may form a lower portion of the dehumidifier 10. A plurality of wheels 16 may be mounted on a bottom of the base 11 to facilitate movement of the dehumidifier 10. The case 12 may be disposed on an upper edge of the base 11, and an upper opening of the case 12 may be shielded by the top cover 13.

[0018] A suction grill 121 may be formed on a back of the case 12 to intake indoor air. A filter module may be mounted in front of the suction grill 121. The suction grill 121 may be defined as a suction port or inlet through which indoor air is suctioned.

[0019] A main discharge port 131, through which dehumidified air may be discharged may be formed on one or a first side of the top cover 13. The main discharge port 131 may be selectively opened or closed by the discharge cover 14.

[0020] A handle 15 may be provided on the other or a second side of the top cover 13. The handle 15 has an n-shape so that a vertical portion thereof may be inserted into the case 12 or pulled out from the case 12. When the vertical portion is fully inserted into the case 12, a horizontal portion of the handle 15, that is, an upper surface of a portion configured to be held by a hand, may be flush with the top cover 13.

[0021] The main discharge port 131 may be disposed at one or a first edge of the top cover 13. The handle 15 may be placed on the other or a second edge of the top cover 13. The main discharge port 131 and the handle 15 may be located on opposite edges. A discharge grill 141 may be mounted on the main discharge port 131.

[0022] A cutout into which the water tank 20 may be inserted may be formed at one edge of the case 12, and the water tank 20 may be detachable from the case 12. When the water tank 20 is mounted on the case 12, an outer surface of the water tank 20 may form a portion of the case 12. The water tank 20 may be located right beneath the main discharge port 131.

[0023] The dehumidification module may include a compressor 31 that compresses a refrigerant, a condenser 32 that condenses the refrigerant that has passed through the compressor 31, an expansion valve that expands the refrigerant that has passed through the condenser 32 into a two-phase refrigerant, and an evaporator 33 that evaporates the refrigerant that has passed through the expansion valve. The dehumidification module may further include a fan module 34 that suctions indoor air through the suction grill 121. The evaporator 33 may be located in front of the suction grill

121, the condenser 32 may be located in front of the evaporator 33, and the fan module 34 may be located in front of the condenser 32.

[0024] The condenser 32 and the evaporator 33 may be seated on an upper surface of a drain pan 35. Condensed water flowing down from a surface of the evaporator 33 may flow along the drain pan 35 and collected in the water tank 20.

[0025] An outlet of the fan module 34 may be connected to the main discharge port 131.

[0026] The air introduced into the case 12 through the suction grill 121 may be dehumidified while passing through the evaporator 33, and moisture contained in the air condensed. As a result, the incoming air becomes low-temperature and dry as it passes through the evaporator 33.

[0027] The low-temperature dry air absorbs heat while passing through the condenser 32, and as a result, is changed to a high-temperature dry state and is then discharged back into the room through the main discharge port 131. The term "high temperature air" may refer to air with a temperature slightly higher than a temperature of an indoor space in which the dehumidifier 10 is installed.

[0028] A control box 17 may be accommodated inside of the case 12, to control operation of the dehumidification module and various drive means.

[0029] A sub discharge port 122 through which dehumidified air may be discharged may be further formed in the case 12. The sub discharge port 122 may be selectively opened or closed by a damper module 40.

[0030] When the sub discharge port 122 is open, the main discharge port 131 may be closed, and when the sub discharge port 122 is closed, the main discharge port 131 may be open. Accordingly, the dehumidified air may be selectively discharged only through either the main discharge port 131 or the sub discharge port 122.

[0031] The sub discharge port 122 may be formed on a front surface or a rear surface of the case 12. The sub discharge port 122 may be disposed at an upper side of the fan module 34. For example, the sub discharge port 122 may be formed on the rear surface of the case 12 and may be located above the suction grill 121.

[0032] The dehumidifier 10 may further include the damper module 40 that selectively opens or closes the sub discharge port 122. The damper module 40 may be installed inside of the sub discharge port 122 and may open or close the sub discharge port 122. The damper module 40 may operate in a direction to close the sub discharge port 122.

[0033] FIG. 4 is a perspective view of a base of a housing of the dehumidifier according to an embodiment. FIG. 5 is a longitudinal cross-sectional view of the base, taken along V-V in FIG. 4.

[0034] Referring to FIGS. 4 and 5, an upper surface of the base 11 of the dehumidifier 10 according to an embodiment may include a water tank seating portion 111, a compressor seating portion 112, and a control box seating portion 113. An external air inlet 114 may be formed on or at one side of the base 11 corresponding to the control box seating portion 113. External air may refer to air outside of the dehumidifier 10, that is, indoor air of an indoor space in which the humidifier 10 is disposed.

[0035] The external air inlet 114 may be provided in the shape of a hole that penetrates the base 11 or an external air inlet sleeve with an open upper surface as shown in the drawings. The external air inlet 114 may be located right

below the control box 17, and may be formed at a position spaced apart from a bottom surface of the control box 17.

[0036] FIGS. 6 to 8 are schematic diagrams showing a flow of indoor air flowing in through an external air inlet of the base. Referring to FIG. 6, some of the indoor air flowing into the dehumidifier 10 through the external air inlet 114 flows into the control box 17 through an opening formed on the bottom surface of the control box 17, and a remainder flows upwards along bottom and side surfaces of the control box 17. That is, the air cools the control box 17 while flowing up along a narrow space formed between the case 12 and the control box 17. Additionally, a circuit board 173 on which a number of electrical components are mounted may be installed inside of the control box 17, and in particular, a heat sink 171 may be mounted on the heating component 172.

[0037] Additionally, the heat sink 171 may penetrate the control box 17 and protrude out of the control box 17. That is, the heat sink 171 may be exposed to the narrow space between the case 12 and the control box 17 and may directly exchange heat with air that flows upwards after flowing into the external air inlet 114. Indoor air flowing in through the external air inlet 114 may cool the heat sink 171 as it flows upwards along the space formed between the control box 17 and the case 12.

[0038] Referring to FIG. 7, the indoor air that flows upwards while cooling the control box 17 may flow towards the drain pan 35 and flow into a gap between the fan module 34 and the drain pan 35. The drain pan 35 may include a first portion 351 that laterally extends so that a fan housing of the fan module 34 may be seated thereon, and second portions 352 that extend downward from a bottom of the first portion 351 to be located on the base 11.

[0039] The second portions 352 may extend from one or a first edge of the first portion 351 and a point spaced apart from the first edge towards the other or a second side, respectively. The control box 17 may be coupled to a second portion that extends from the first edge of the first portion 351 among the second portions 352, and the second portion that extends from the point spaced from the first edge towards the second side is configured to surround a side surface of the water tank 20.

[0040] In addition, an air guide portion (air guide) 353 may be formed at one edge of the first portion 351, in particular, at the edge of the first portion 351 corresponding to an upper end of the second portion 352 on which the control box 17 is mounted. Air flowing up along the surface of the control box 17 flows up towards the fan module 34 through the air guide portion 353.

[0041] Referring to FIG. 8, the fan module 34 may include the fan housing and a fan accommodated inside of the fan housing. A dripping hole 341 may be formed at a lower end of the fan housing. During the dehumidification process, condensed water condensed on a surface of the fan and an inner surface of the fan housing may be discharged to the drain pan 35 through the dripping hole 341.

[0042] The air that flows upward while cooling the control box 17 flows into the fan housing through the dripping hole 341. The air introduced into the fan housing may be discharged back into the room through the main discharge port 131 while evaporating some of the condensate condensed on the fan and the fan housing, when the fan rotates.

[0043] By forming the dripping hole 341, when the fan rotates, negative pressure is created outside of the fan

housing, allowing indoor air to flow into the dehumidifier 10 through the external air inlet 114. In addition, the indoor air flowing into the dehumidifier 10 through the external air inlet 114 cools the control box 17 as it flows upward, and absorbs heat while passing through the control box 17, increasing its temperature.

[0044] In addition, the indoor air with an increased temperature flows into the fan module 34 through the dripping hole 341, and also performs the function of evaporating condensation water formed inside of the fan module 34. The air flowing into the fan module 34 through the dripping hole 341 dries the inside of the fan module 34, thereby preventing fungus from growing inside of the fan module 34.

[0045] Embodiments disclosed herein provide a dehumidifier that may include a base with an external air intake on one side thereof; a case erected on an edge of an upper surface of the base and having a suction grille formed on a rear surface thereof; a top cover placed on an upper surface of the case and having a discharge port; a discharge cover that opens or closes the discharge port; a drain pan including a first part or portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second part or portion that extends from a lower surface of the first part and placed on the base; a water tank placed on one side of the upper surface of the base and that stores condensed water generated during a dehumidification process; a compressor placed on one side of the upper surface of the base spaced apart from the water tank; a fan module placed on the first portion and having an outlet that communicates with the discharge port of the top cover; a condenser mounted on the first portion to be located on an intake side of the fan module; an evaporator mounted on the first portion to be located behind the condenser; and a control box coupled to the second part and erected at a point right above the external air intake. The fan module may include a fan housing placed on the first portion; and a fan accommodated in the fan housing. A dripping hole may be formed in the fan housing.

[0046] The external air inlet may be formed on an upper surface of a sleeve that extends upward from the base. A separated space may be formed between the case and the control box so that indoor air flowing into the case through the external air inlet may flow upward while contacting the control box. An air guide portion may be formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box may be guided to an upper side of the first portion. The dripping hole may be formed at a lower edge of the fan housing.

[0047] According to embodiments disclosed herein, a hole through which indoor air flows may be formed at a bottom of the dehumidifier where the control box is located, such that indoor air flowing in through the hole may flow upward to cool the control box, and thereafter flow into the fan module through the dripping hole formed in the fan module. As a result, electrical components that generate a lot of heat may be cooled quickly, thereby preventing overheating of the control box.

[0048] It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present.

As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0049] It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

[0050] Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0051] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0052] Embodiments are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0053] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0054] Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such phrases in various places in the specification are not

necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

[0055] Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

What is claimed is:

1. A dehumidifier, comprising:
 - a base with an external air intake on one side thereof;
 - a case disposed on an upper surface of the base and having a suction grille formed on a rear surface thereof;
 - a top cover disposed on an upper surface of the case and having a discharge port;
 - a discharge cover that opens and closes the discharge port;
 - a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends from a lower surface of the first portion and disposed on the base;
 - a water tank disposed on the upper surface of the base and that stores condensed water generated during a dehumidification process;
 - a compressor disposed on the upper surface of the base spaced apart from the water tank;
 - a fan module disposed on the first portion and having an outlet that communicates with the discharge port of the top cover;
 - a condenser mounted on the first portion at an intake side of the fan module;
 - an evaporator mounted on the first portion behind the condenser; and
 - a control box coupled to the second portion and disposed directly above the external air intake, wherein the fan module includes:
 - a fan housing disposed on the first portion;
 - a fan accommodated in the fan housing, and wherein a dripping hole is formed in the fan housing.
2. The humidifier according to claim 1, wherein the external air intake is formed at an upper surface of a sleeve that extends upward from the base.
3. The humidifier according to claim 2, wherein a space is formed between the case and the control box so that indoor air flowing into the case through the external air intake flows upward along the control box.
4. The humidifier according to claim 3, wherein an air guide is formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box is guided to an upper side of the first portion.
5. The humidifier according to claim 4, wherein the dripping hole is formed at a lower edge of the fan housing.
6. A dehumidifier, comprising:
 - a base with an external air intake;
 - a case disposed on the base and having a suction inlet formed on a surface thereof;
 - a top cover disposed on the case and having a discharge outlet;
 - a discharge cover that opens and closes the discharge outlet;
 - a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends downward from the first portion to the base;
 - a water tank disposed on the base and that stores condensed water generated during a dehumidification process;
 - a compressor disposed on the base spaced apart from the water tank;
 - a fan module disposed on the first portion and having an outlet that communicates with the discharge outlet of the top cover;
 - a condenser mounted on the first portion at an intake side of the fan module;
 - an evaporator mounted on the first portion behind the condenser; and
 - a control box coupled to the second portion and disposed directly above the external air intake, wherein the fan module includes:
 - a fan housing disposed on the first portion;
 - a fan accommodated in the fan housing, wherein a space is formed between the case and the control box so that indoor air flowing into the case through the external air intake flows upward along the control box.
7. The humidifier according to claim 6, wherein the external air intake is formed at an upper surface of a sleeve that extends upward from the base.
8. The humidifier according to claim 6, wherein an air guide is formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box is guided to an upper side of the first portion.
9. The humidifier according to claim 6, wherein the dripping hole is formed at a lower edge of the fan housing.
10. A dehumidifier, comprising:
 - a base with an external air intake that extends upward therefrom;
 - a case disposed on the base and having a suction inlet formed on a surface thereof;
 - a top cover disposed on an upper surface of the case and having a discharge outlet;
 - a discharge cover that opens and closes the discharge outlet;
 - a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends downward from a lower surface of the first portion to the base;
 - a water tank disposed on the base and that stores condensed water generated during a dehumidification process;
 - a compressor disposed on the base spaced apart from the water tank;
 - a fan module disposed on the first portion and having an outlet that communicates with the discharge outlet of the top cover;

a condenser mounted on the first portion at an intake side of the fan module;
an evaporator mounted on the first portion behind the condenser; and
a control box coupled to the second portion and disposed directly above the external air intake, wherein the fan module includes:
a fan housing disposed on the first portion;
a fan accommodated in the fan housing, wherein a dripping hole is formed in the fan housing through which condensed water condensed on a surface of the fan and an inner surface of the fan housing is discharged to the drain pan, and wherein a space is formed between the case and the control box so that indoor air flowing into the case through the external air intake flows upward along the control box.

11. The humidifier according to claim **10**, wherein the external air intake includes a sleeve that extends upward from the base.

12. The humidifier according to claim **10**, wherein an air guide is formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box is guided to an upper side of the first portion.

13. The humidifier according to claim **10**, wherein the dripping hole is formed at a lower edge of the fan housing.

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